

Multiple Choice

1. **Answer:** A

Options: A) the floor; B) the pull of gravity; C) the air friction on the ball; D) the person that throws the ball

Note: The floor provides the upward force through the normal force, which acts in opposition to the downward force of gravity when the ball makes contact with it, allowing the ball to bounce back upward.

2. **Answer:** D

Options: A) Eye color; B) Hair color; C) Height; D) Verbal accent

Note: Verbal accent is an acquired human characteristic because it develops as individuals are exposed to and learn the speech patterns and pronunciations of the language spoken in their environment, rather than being genetically inherited.

3. **Answer:** B

Options: A) height.; B) mass.; C) shape.; D) weight.

Note: The amount of gravitational attraction is directly proportional to the mass of the objects involved, as described by Newton's law of universal gravitation, which states that the force of gravity increases with greater mass.

4. **Answer:** D

Options: A) hibernate for the season; B) migrate to warmer climates; C) change their body covering; D) move to seek shelter

Note: The correct answer is "move to seek shelter" because many animals instinctively seek protection from rain to avoid getting wet and to maintain their body temperature, demonstrating their adaptive behavior for survival.

5. **Answer:** C

Options: A) Turn away after two or three minutes.; B) Look at the Sun through a long telescope.; C) Cast an image through a pinhole onto a screen.; D) Blink often until your eyes get used to the light.

Note: Casting an image through a pinhole onto a screen is the safest method to watch a solar eclipse because it allows viewers to see the image of the Sun indirectly, preventing harmful exposure to the intense sunlight that can damage the eyes.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: When a bird drops a seed in soil, it provides the seed with a suitable environment for germination and growth, thus enhancing its chances of developing into a new plant.

7. **Answer:** False

Options: A) True; B) False

Note: A simple machine does not change the weight of an object; instead, it changes the way the force is applied to move the object, making it easier to lift without altering its mass.

8. **Answer:** False

Options: A) True; B) False

Note: A rabbit's fur turning white in winter is an adaptation to seasonal change that provides camouflage in snowy environments, not a response to habitat destruction.

9. **Answer:** False

Options: A) True; B) False

Note: The statement is false because a boulder sitting on a cliff possesses potential energy due to its height, but it does not represent a conversion of kinetic energy to potential energy, as kinetic energy is associated with motion, which the boulder does not have while stationary.

10. **Answer:** False

Options: A) True; B) False

Note: Coal is classified as a non-renewable resource because it forms over millions of years from the decomposition of ancient organic matter, making it impossible to replenish on a human timescale.

Long Form Response

11. **Answer:** Farmers choose which crops to grow based on several key factors, including climate, soil type, market demand, and available resources. For example, certain crops thrive in specific climates and soil conditions, which can determine a farmer's success. Market demand influences farmers to grow crops that will sell well, ensuring economic viability. Additionally, the availability of water and technology can impact agricultural practices, pushing farmers to adopt more sustainable methods like crop rotation or organic farming to maintain soil health and reduce environmental impact. Overall, these factors not only shape the farmers' choices but also play a crucial role in promoting sustainable agricultural practices.

Note: The answer correctly identifies and explains that a farmer's crop choices are influenced by climate, soil type, market demand, and resource availability, highlighting how these factors affect agricultural practices and sustainability.

12. **Answer:** Temperature changes along coastlines are generally less extreme than those experienced further inland due to the moderating influence of the ocean. Water has a high specific heat capacity, meaning it can absorb and store large amounts of heat without significantly changing temperature. As a result, coastal areas benefit from the ocean's ability to maintain more stable temperatures, preventing extreme highs and lows. In contrast, inland areas, which are further away from large bodies of water,

experience more drastic temperature variations because land heats up and cools down more quickly than water. This combination of oceanic influence and land behavior leads to milder coastal climates.

Note: correct because it highlights that the ocean's high specific heat capacity allows it to absorb and regulate temperature changes, resulting in milder temperature variations along coastlines compared to the more extreme temperature fluctuations experienced further inland.

End of Answer Key